ArmorGalv®

SPENCER INDUSTRIES

LCAC Tie Downs

Document #D04

Spencer INDUSTRIES INC.
SUPPORTING THE MILITARY FOR OVER 40 YEARS

MEGA RUST 2013

SPENCER SPOTLIGHT

Spencer Industries Inc. pioneered the application of ASTM A 1059 Zinc Alloy Thermo-Diffusion Coatings --trade named **ArmorGalv®** -- for the US Navy in 2008 as part of the Intra-Connector Material Handling (ICMH) System, an ONR-sponsored project managed by SCRA. Through the ICMH project, NSWC-PCD installed **ArmorGalv®** protected chains and tensioners on LCAC-66 operating out of Panama City, Florida in July 2010. The chains and tensioners have remained corrosion-free over the three-year period since initial installation.

The inner pair of cargo lashings, manufactured by Spencer, were coated with **ArmorGalv®**. They exhibit no signs of rust after three years in the highly corrosive Gulf environment. The outer pair of legacy lashings, using standard materials and treatment, have significantly corroded over the same period.

Photo courtesy of Thomas Whelan, Mechanical Engineer, ACV Branch, NSWC-PCD thomas.whelan@navy.mil

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Current Projects

USS NEW YORK (LPD 21)

In October 2011, as a demonstration project sponsored by SEA-21/SURFMEPP, **Spencer Industries** removed six curb plates from LPD 21's flight deck. Each panel had all paint removed and **ArmorGalv®** coating was then applied. The panels were then returned to LPD 21 for re-installation and painting. Over the subsequent six month deployment, there was no evidence of corrosion on any of the six plates, while corrosion continued to propagate on the remaining curb plates.

Based on the success of the demonstration, in May 2013 all untreated curb plates were removed from the flight deck and paint was removed by **Amee Bay** using the dry ice blasting alternative to traditional sand blasting. The plates were then **ArmorGalv®** coated with a tri-top sealer by **Spencer Industries,** and returned to Amee Bay for re-painting and installation. The project is scheduled for completion by the end of June 2013.

The differences between **ArmorGalv®** coated and uncoated curb plates are readily apparent in these photographs. After six months deployment, the **ArmorGalv®** coated plates remain paint-fresh beside untreated plates that show excessive rusting.
Photos - NAVSEA

RE-USABLE, NON-CORRODING LOCKNUTS

Spencer Industries and partner **Security Locknut Inc.** co-developed a high-performance, non-corroding, re-usable steel locknut for use in a wide range of military, transportation and manufacturing equipment.

Current Projects

USAV BRISTOE STATION (LCU 2006) and USAV NEW ORLEANS (LCU 2031)
Spencer Industries, in collaboration with the U.S. Army Tank Automotive Command Watercraft Inspection Branch, U.S. Army Logistics Innovation Agency and partner **Metal Trades Inc.**, initiated an evaluation of the zinc diffusion process for improved corrosion resistance on the US Army's Runnymede class large landing craft. In February 2013, a total of 13 anchor chains, two bow chains and five detachable links were sand-blasted and **ArmorGalv®** coated by Spencer. The chains and links were then re-painted by Metal Trades, and re-installed on LCU 2031. **ArmorGalv®** coated anchor chains and components of the steam boat ratchet assembly will be installed on LCU 2006 during upcoming maintenance in June/July 2013.

USS FREEDOM (LCS 1)
In May 2010, **Spencer Industries**, at the request of VADM McCoy of NAVSEA, performed a corrosion inspection of LCS 1 to identify parts that could be treated with **ArmorGalv®** for enhanced corrosion protection and reduced maintenance costs. For one potential product family, Spencer teamed with **Samtan Engineering Corp.** to develop 50-year clip-type and double-leg style pipe hangers. Samples of both types were fabricated, coated with **ArmorGalv®**, and will be provided to the Navy for evaluation.

Current Projects

CARGO TIEDOWNS

Spencer Industries is in the process of developing new cargo tiedown devices for amphibious ships and landing craft that are lighter, safer and far less prone to corrosion, including corrosion propagation on the decks. For LCACs ACU 4 and ACU 5, Spencer provided 100 tie-Downs coated with **ArmorGalv®**, which have now been deployed for over one year with no evidence of corrosion.

Under a MARCORSYSCOM SBIR (POCs Lou Carl louis.carl@usmc.mil and Maurice Johnson maurice.e.johnson@usmc.mil), Spencer has developed and demonstrated a replacement tiedown that is 30% lighter, corrosion-resistant, safer and easier to use than legacy devices. Under the upcoming SBIR Phase II, lab and field testing will be performed.

NEW AND EXCITING…

A current R & D project is showing important and potentially long-reaching results. Aluminum parts were coated with **ArmorGalv®**, resulting in a part that is harder and more resistant to corrosion than that of a hard anodize.

For those applications that cannot be **ArmorGalv®** coated in the factory, we now offer an HHA-approved water-based paint known as **ERX™ Ecological Rust eXterminating.**

ERX™, when applied over a corroded surface after a water jet cleaning, bonds chemically to the surface by converting the rust (iron oxide) to an Iron Phosphate – a highly corrosion resistant compound. **ERX™** is currently being used on at-sea oil platforms with excellent results.

Partner with Spencer to resolve your toughest corrosion issues.

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